Mechatronics Tutorial Sheet 2: Fourier Transform & Laplace Transform

1. Suppose that the discrete-time LTI systems shown in Figure 1 have frequency responses:

$$G(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega} + \frac{1}{4}e^{-2j\omega}}, \qquad H(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + \frac{1}{2}e^{-j\omega}}$$

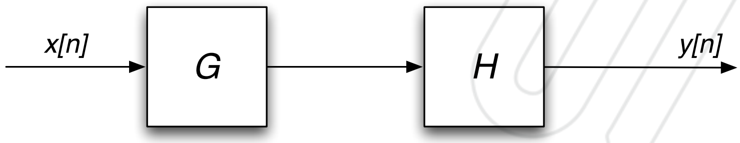

Figure 1

a) Find a difference equation relating $x[n]$ to $y[n]$.

b) Find the impulse response of the combined system.

a) $y[n] = x[n] * G[n] * H[n]$

$$\therefore \frac{Y(e^{jw})}{X(e^{jw})} = G(e^{jw})H(e^{jw}) = \frac{2 - e^{-jw}}{(1 - \frac{1}{2}e^{-jw} + \frac{1}{4}e^{-2jw}) + (\frac{1}{2}e^{-jw} - \frac{1}{4}e^{-2jw} + \frac{1}{8}e^{-3jw})}$$

$$= \frac{2 - e^{-jw}}{1 + \frac{1}{8}e^{-3jw}}$$

$$\therefore (1 + \frac{1}{8}e^{-3jw})Y(e^{jw}) = (2 - e^{-jw})X(e^{jw})$$

$$\therefore y[n] + \frac{1}{8}y[n-3] = 2x[n] - x[n-1]$$

b) for $x[n] = \delta[n] = \begin{cases} 1 & , n = 0 \\ 0 & , \text{otherwise} \end{cases}$

assume that causuality, $y[n] = 0$ for $n < 0$, therefore,

$y[0] + 0 = 2 - 0 = 2 \qquad \therefore y[0] = 2$

$y[1] + 0 = 0 - 1 = -1 \qquad \therefore y[1] = -1$

$y[2] + 0 = 0 \qquad \therefore y[2] = 0$

$y[3] + \frac{1}{8} y[0] = 0 \qquad \therefore y[3] = -\frac{1}{8} y[0] = -\frac{1}{4}$

$y[4] + \frac{1}{8} y[1] = 0 \qquad \therefore y[4] = -\frac{1}{8} y[1] = \frac{1}{8}$

……

Therefore, for $k = 0, 1, 2, 3 \ldots$

$$\begin{cases} y[3k] = (-\frac{1}{8})^k y[0] = 2(-\frac{1}{8})^k \\ y[3k+1] = (-\frac{1}{8})^k y[1] = -(-\frac{1}{8})^k \\ y[3k+2] = 0 \end{cases}$$

2. For each of the following systems, plot the poles and zeros and determine whether or not the system is stable.

a) $H(s) = \frac{1}{s}$

b) $H(s) = \frac{1-s}{s^2-9}$

c) $H(s) = \frac{s}{s+3}$

d) $H(s) = \frac{3}{(s+1)(s+2)}$

e) $H(s) = \frac{s-2}{s+2}$

f) $H(s) = \frac{s+2}{s-2}$

g) $H(s) = \frac{1-s}{s^2+s+1}$

h) $H(s) = \frac{s^2+s+1}{(s+4)(s^2-s+1)}$

	zeros (s =)	poles (s =)	stable
a)	-	0	-
b)	1	± 3	✗
c)	0	-3	✓
d)	-	-1 or -2	✓
e)	2	-2	✓
f)	-2	2	✗
g)	1	$-\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$	✓
h)	$-\frac{1}{2} \pm \sqrt{3} i$	-4 or $\frac{1}{2} \pm \frac{\sqrt{3}}{2} i$	✗

3. Suppose that $H(s)$ is the transfer function of a linear system with input $x(t)$ and output $y(t)$. Find $y(t)$ if:

a) $H(s)$ is as in (2c), and $x(t)$ is an impulse.

b) $H(s)$ is as in (2d), and $x(t)$ is a unit step.

c) $H(s)$ is as in (2g), and $x(t)$ is a unit step.

a) $H(s) = \frac{s}{s+3}$, $x(t) = \delta(t)$

$\therefore X(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-s0} = 1$

$\therefore Y(s) = H(s) \cdot X(s) = \frac{s}{s+3}$

let $y(t) = \frac{d}{dt} z(t)$ $\therefore Y(s) = sZ(s)$ for $z(0) = 0$

$\therefore Z(s) = \frac{1}{s+3}$

$\therefore z(t) = e^{-3t}$

$\therefore y(t) = \frac{d}{dt} z(t) = -3e^{-3t}$ for $t \geq 0$ (causuality)

b) for $t \geq 0$,

$H(s) = \frac{3}{(s+1)(s+2)}$, $x(t) = u(t) = \begin{cases} 1 & \text{for } t \geq 0 \\ 0 & \text{for } t = 0 \end{cases}$

$\therefore X(s) = \int_0^{\infty} u(t) e^{-st} dt = \int_0^{\infty} e^{-st} dt = -\frac{1}{s} [e^{-st}]_0^{\infty} = \frac{1}{s}$

$\therefore Y(s) = X(s) \cdot H(s) = \frac{3}{s(s+1)(s+2)}$

$= \frac{A(s^2+3s+2)}{s(s+1)(s+2)} + \frac{B(s^2+2s)}{s(s+1)(s+2)} + \frac{C(s^2+s)}{s(s+2)(s+1)}$

$= \frac{3}{2} \frac{1}{s} - 3 \frac{1}{s+1} + \frac{3}{2} \frac{1}{s+2}$

$\therefore y(t) = \frac{3}{2} - 3e^{-t} + \frac{3}{2} e^{-2t}$

c) for $t \geq 0$,

$$H(s) = \frac{1-s}{s^2+s+1} \quad \text{and} \quad X(s) = \frac{1}{s}$$

$$\therefore Y(s) = H(s) \cdot X(s) = \frac{1-s}{s^3+s^2+s}$$

$$= \frac{A(s^2+s+1)}{s(s^2+s+1)} + \frac{Bs}{s(s^2+s+1)}$$

$$= \frac{1}{s} - \frac{s+2}{s^2+s+1}$$

$$= \frac{1}{s} - \frac{s}{s^2+s+1} - \frac{2}{s^2+s+1}$$

Taking the inverse Laplace transform

$$y(t) = 1 - s\frac{1}{2\pi j}\int_{6-j\infty}^{6+j\infty}\left(\frac{1}{s^2+s+1}\right)e^{st}\,ds - 2\frac{1}{2\pi j}\int_{6-j\infty}^{6+j\infty}\left(\frac{1}{s^2+s+1}\right)e^{st}\,ds$$

$$= 1 - \frac{d}{dt}\left(\frac{e^{-\frac{1}{2}t}}{\sqrt{1-(\frac{1}{2})^2}}\sin\left(\sqrt{1-(\tfrac{1}{2})^2}\,t\right)\right) - 2\left(\frac{e^{-\frac{1}{2}t}}{\sqrt{1-(\frac{1}{2})^2}}\sin\left(\sqrt{1-(\tfrac{1}{2})^2}\,t\right)\right)$$

$$= 1 - \frac{d}{dt}\left[\frac{2}{3}\sqrt{3}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right)\right] - \frac{4}{3}\sqrt{3}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

$$= 1 - \sqrt{3}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right) + e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{3}}{2}t\right)$$

4. a) Suppose that the signal $x(t)$ has Laplace transform $X(s)$. Show that

$$x(t-T) \xleftrightarrow{\mathcal{L}} e^{-Ts}X(s)$$

for $y(t) = x(t-T), \; t \geq 0$ $\quad \therefore t - T \geq 0, \; t \geq T$

$$Y(s) = \int_0^{\infty} y(t)\, e^{-st}\, dt = \int_0^{\infty} x(t-T)\, e^{-st}\, dt$$

Let $\tau = t - T \quad \therefore t = \tau + T$

$$\therefore Y(s) = \int_{0-T}^{\infty-T} x(\tau)\, e^{-s(\tau+T)}\, d\tau$$

$$= e^{-sT} \int_T^{\infty} x(\tau)\, e^{-s\tau}\, d\tau$$

$$= e^{-sT} X(s)$$

5. a) Find the transfer function $\frac{Y(s)}{X(s)}$ for the system shown in Figure 5.1.

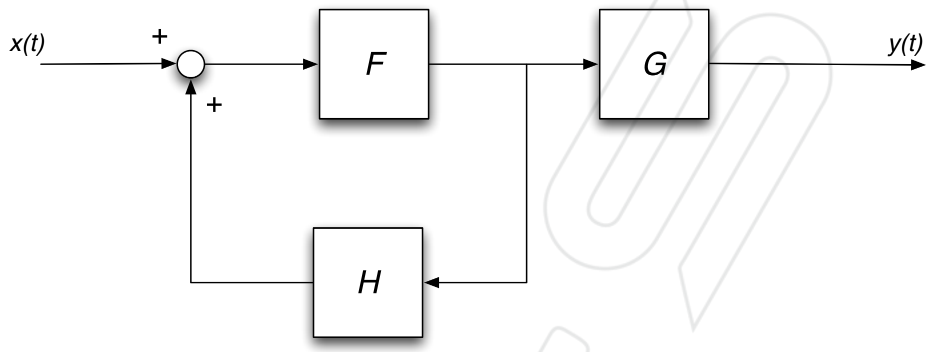

Figure 5.1

$$\frac{Y(s)}{X(s)} = \frac{F(s)G(s)}{1 - F(s)H(s)}$$

b) Find the transfer function $\frac{Y(s)}{X(s)}$ for the system shown in Figure 5.2.

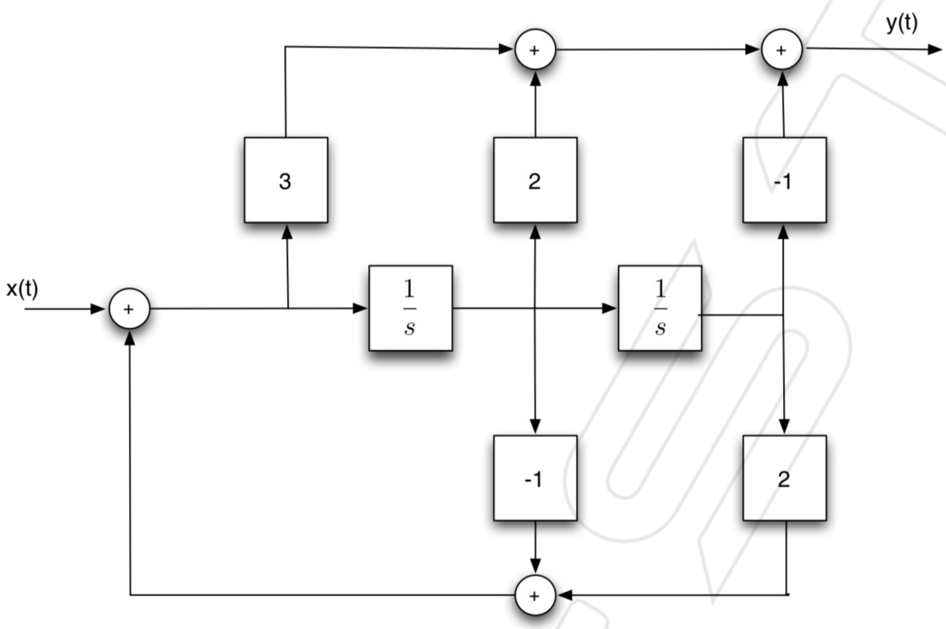

Figure 5.2

$$\frac{Y(s)}{X(s)} = \frac{3 + (\frac{1}{s} \cdot 2) + [\frac{1}{s} \cdot \frac{1}{s} \cdot (-1)]}{1 - [\frac{1}{s} \cdot (-1) + \frac{1}{s} \cdot \frac{1}{s} \cdot 2]}$$

$$= \frac{-\frac{1}{s^2} + \frac{2}{s} + 3}{-\frac{2}{s^2} + \frac{1}{s} + 1}$$

$$= \frac{-1 + 2s + 3s^2}{-2 + s + s^2}$$

c) In the system shown in Figure 5.3, find

i) The transfer function from input to tracking error $\frac{e(s)}{x(s)}$

ii) The transfer function from measurement noise to output $\frac{y(s)}{n(s)}$

iii) The transfer function from disturbance to output $\frac{y(s)}{d(s)}$

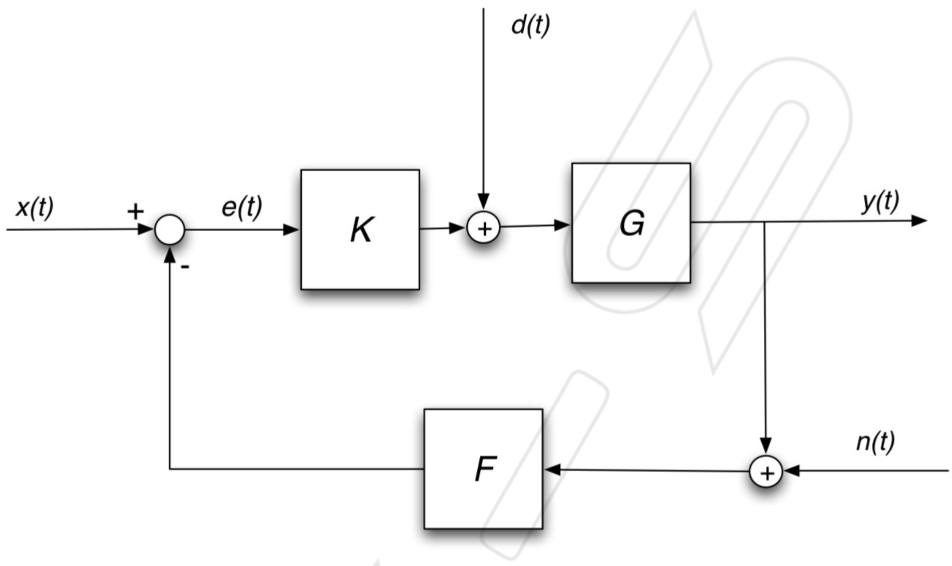

Figure 5.3

i) $$\frac{e(s)}{x(s)} = \frac{\cancel{-K(s)G(s)F(s)}\ 1}{1 + K(s)G(s)F(s)}$$

ii) $$\frac{y(s)}{n(s)} = \frac{-K(s)G(s)F(s)}{1 + K(s)G(s)F(s)}$$

iii) $$\frac{y(s)}{d(s)} = \frac{G(s)}{1 + K(s)G(s)F(s)}$$

6. Suppose that in Figure 5.3 the following transfer functions are given:

$$G(s) = \frac{s+1}{(s-2)(s+3)}, \qquad F(s) = 1, \qquad K(s) = K,$$

where K is a real number. Find the range of values of K for which the system is stable. Consider the input to be $x(t)$ and the output to be $y(t)$.

$$\frac{y(s)}{x(s)} = \frac{K(s)\,G(s)}{1 + K(s)\,G(s)\,F(s)}$$

$$= \frac{K(s+1)}{(s-2)(s+3) + K(s+1)}$$

$$= \frac{K(s+1)}{s^2 + (1+K)s + (K-6)}$$

for pole. $s^2 + (1+K)s + (K-6)$

$$\therefore s = \frac{-(1+K) \pm \sqrt{(1+K)^2 - 4(K-6)}}{2}$$

for $Re(s) < 0$.

$k > 6$